

APPLIED DATA STRUCTURES & ALGORITHMS LAB (BCS-553)

B.Tech CSE 5th SEMESTER

SUBMMITED TO:

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Lab Assignment 7: Practice Set

Experiment No. 7.1 - Search for 52 with a loop

Experiment Title:

Search for 52 with a loop

Aim:

To search for the value 52 in an array and print its index or 'Not found'.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

This practice set combines the earlier ideas: scanning an array with a loop to search, sorting with bubble sort, accumulating a product with a loop, and filtering values with an if statement. A value is even when value % 2 is 0.

Algorithm / Procedure:

1. Set index = -1.
2. Loop over the array; if a[i] == 52, store i in index and break.
3. If index != -1 print Found at index followed by the index.
4. Otherwise print Not found.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 7, Task 1
// Question: Search for the value 52 in the array {45, 52, 80} with a
// loop. If 52 is there, print "Found at index" followed by a space
// and the index. If it is not there, print "Not found".

public class Main {
    public static void main(String[] args) {
        int[] a = {45, 52, 80};
        int index = -1;

        for (int i = 0; i < a.length; i++) {
            if (a[i] == 52) {
                index = i;
                break;
            }
        }

        if (index != -1) {
```

```
            System.out.println("Found at index " + index);
        } else {
            System.out.println("Not found");
        }
    }
}
```

Input:

Array {45, 52, 80}, value = 52.

Output:

```
Found at index 1
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The value 52 was found at index 1.

Viva-Voce Questions:

- What does the break statement do?
- Why is index initialised to -1?
- What is printed if the value is absent?

Experiment No. 7.2 - Bubble sort of {3, 1, 2}

Experiment Title:

Bubble sort of {3, 1, 2}

Aim:

To sort the array {3, 1, 2} using bubble sort.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

This practice set combines the earlier ideas: scanning an array with a loop to search, sorting with bubble sort, accumulating a product with a loop, and filtering values with an if statement. A value is even when value % 2 is 0.

Algorithm / Procedure:

1. Create the array {3, 1, 2}.
2. Use an outer loop for the passes and an inner loop for the comparisons.
3. Swap neighbours when the left one is larger.
4. Print every value of the sorted array.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 7, Task 2
// Question: Sort the array {3, 1, 2} with bubble sort and print each
// value of the sorted array on its own line.

public class Main {
    public static void main(String[] args) {
        int[] a = {3, 1, 2};

        for (int i = 0; i < a.length - 1; i++) {
            for (int j = 0; j < a.length - 1 - i; j++) {
                if (a[j] > a[j + 1]) {
                    int temp = a[j];
                    a[j] = a[j + 1];
                    a[j + 1] = temp;
                }
            }
        }

        for (int i = 0; i < a.length; i++) {
            System.out.println(a[i]);
        }
    }
}
```

Input:

Fixed array declared in the program: {3, 1, 2}.

Output:

```
1
2
3
```

Time / Space Complexity: Time $O(n^2)$; Space O(1).

Result:

The program was compiled and executed successfully. The array was sorted in ascending order: 1, 2, 3.

Viva-Voce Questions:

- How many passes are needed for 3 elements?
- What is the time complexity of bubble sort?
- Why is a temporary variable used?

Experiment No. 7.3 - Product of array values

Experiment Title:

Product of array values

Aim:

To multiply all the values of an array using a for loop.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

This practice set combines the earlier ideas: scanning an array with a loop to search, sorting with bubble sort, accumulating a product with a loop, and filtering values with an if statement. A value is even when value % 2 is 0.

Algorithm / Procedure:

1. Create the array {2, 3, 4} and set product = 1.
2. Loop over every element.
3. Multiply product by each value.
4. Print product with the label Product.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 7, Task 3
// Question: Use a for loop to multiply all the values of {2, 3, 4}
// and print the product as "Product:" followed by the number.

public class Main {
    public static void main(String[] args) {
        int[] a = {2, 3, 4};
        int product = 1;

        for (int i = 0; i < a.length; i++) {
            product = product * a[i];
        }

        System.out.println("Product: " + product);
    }
}
```

Input:

Fixed array declared in the program: {2, 3, 4}.

Output:

```
Product: 24
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The product of the values (24) was calculated correctly.

Viva-Voce Questions:

- Why is product initialised to 1 and not 0?
- What would happen if it started at 0?
- What is the product of the given array?

Experiment No. 7.4 - Print even values

Experiment Title:

Print even values

Aim:

To print each even value of an array using a loop and an if statement.

Software / Tools Required:

Java JDK, a text editor or IDE (such as IntelliJ IDEA, Eclipse, or VS Code), and a terminal/command prompt.

Theory:

This practice set combines the earlier ideas: scanning an array with a loop to search, sorting with bubble sort, accumulating a product with a loop, and filtering values with an if statement. A value is even when value % 2 is 0.

Algorithm / Procedure:

1. Create the array {1, 2, 3, 4}.
2. Loop over every element.
3. If a[i] % 2 == 0, print a[i].
4. Skip odd values.

Program / Queries / Commands:

```
// Name: Mankirat Singh
// ID: BCS2024542
// Roll No: 24103032571
// Assignment 7, Task 4
// Question: Use a for loop and an if statement to print each even
// value in {1, 2, 3, 4}, one per line. A value is even when value %
// 2 is 0.

public class Main {
    public static void main(String[] args) {
        int[] a = {1, 2, 3, 4};

        for (int i = 0; i < a.length; i++) {
            if (a[i] % 2 == 0) {
                System.out.println(a[i]);
            }
        }
    }
}
```

Input:

Fixed array declared in the program: {1, 2, 3, 4}.

Output:

```
2
4
```

Time / Space Complexity: Time O(n); Space O(1).

Result:

The program was compiled and executed successfully. The even values 2 and 4 were printed.

Viva-Voce Questions:

- What does the % operator do?
- How do you test whether a number is odd?
- What is the output for {1, 2, 3, 4}?